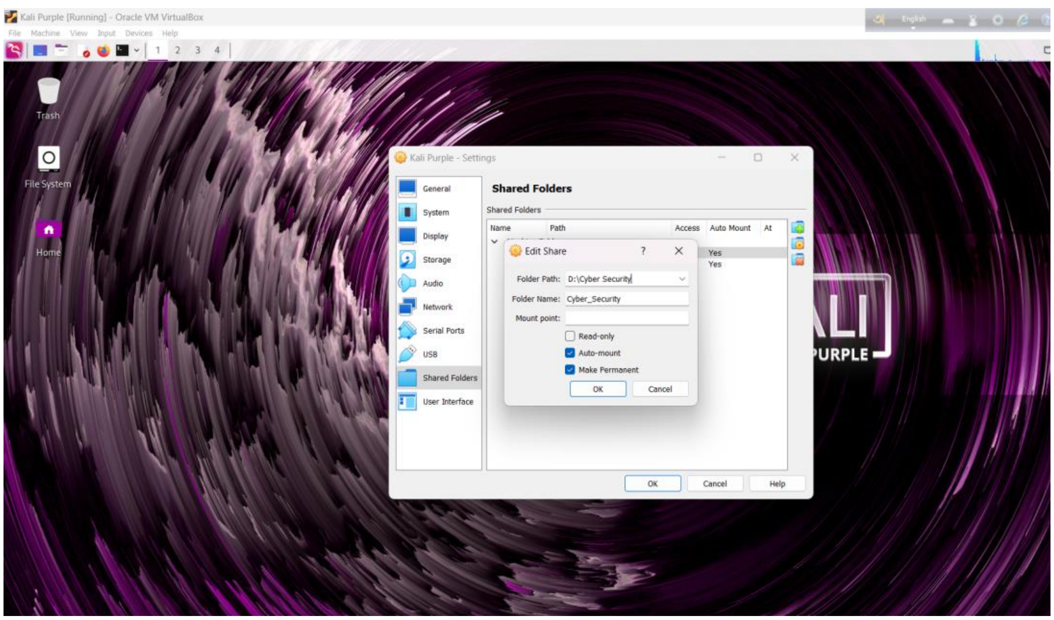
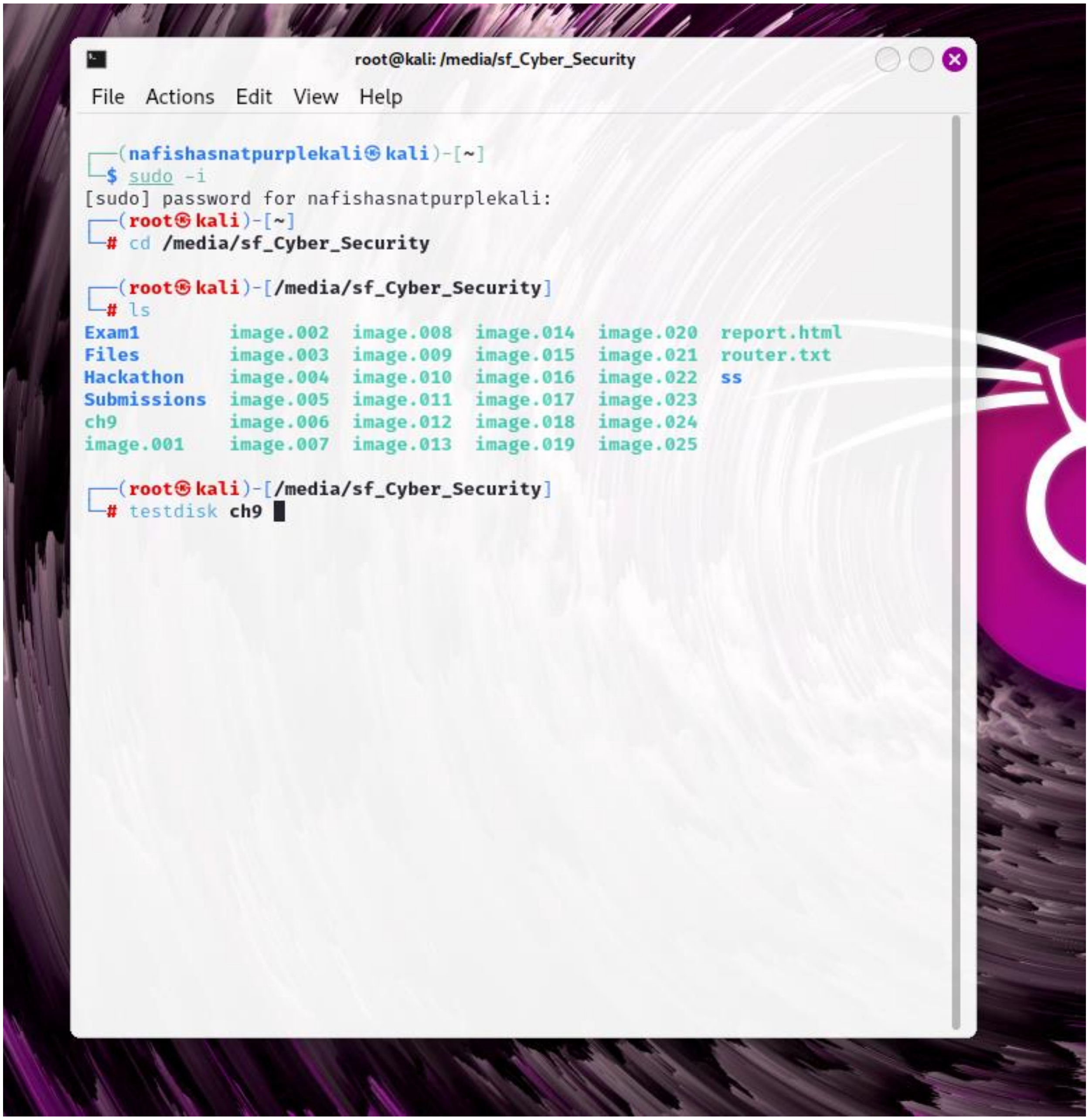
Cyber Security:Class-05
Date: 21 September 2024
Topics : Forensic of an Image

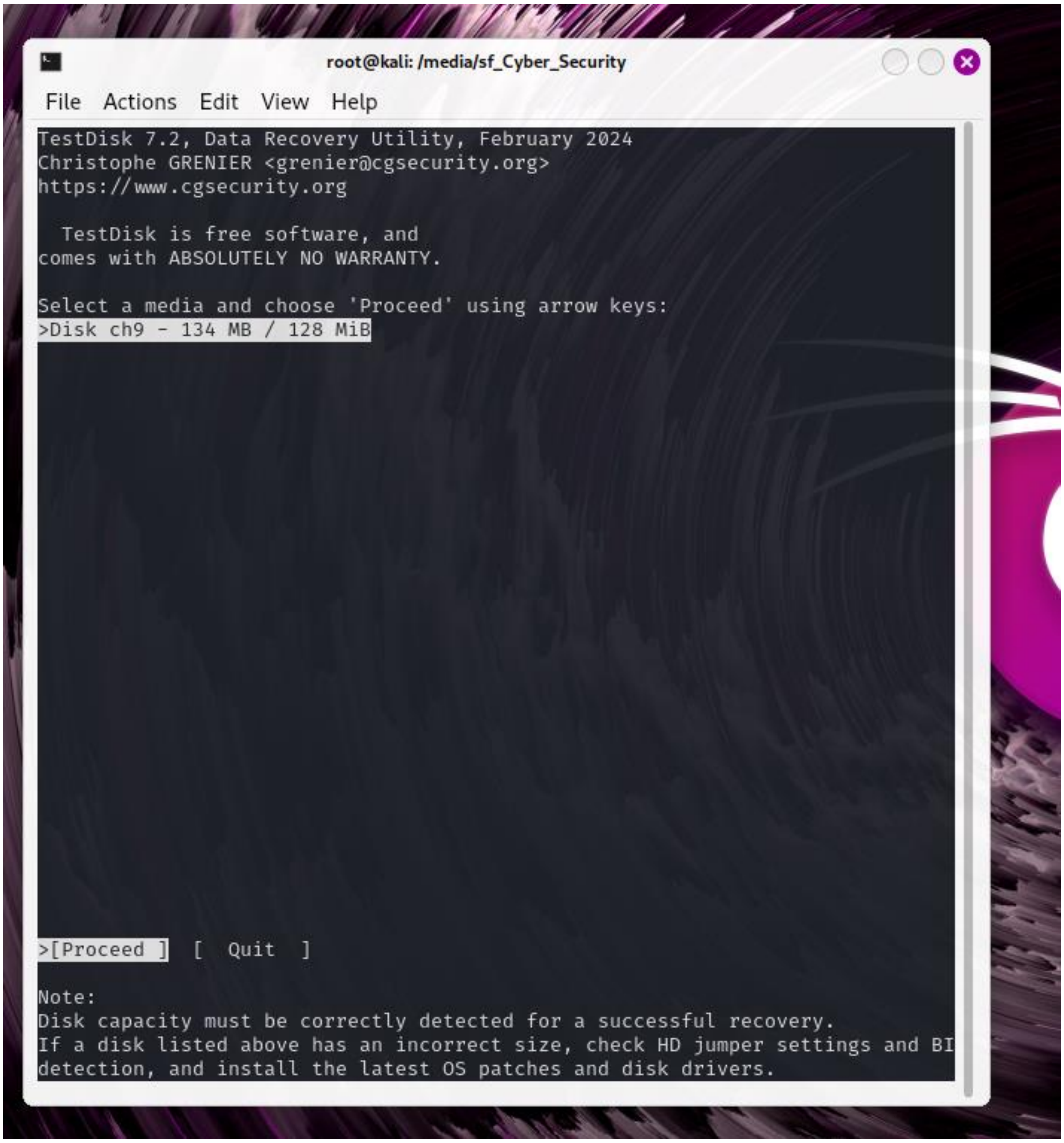
Steps:

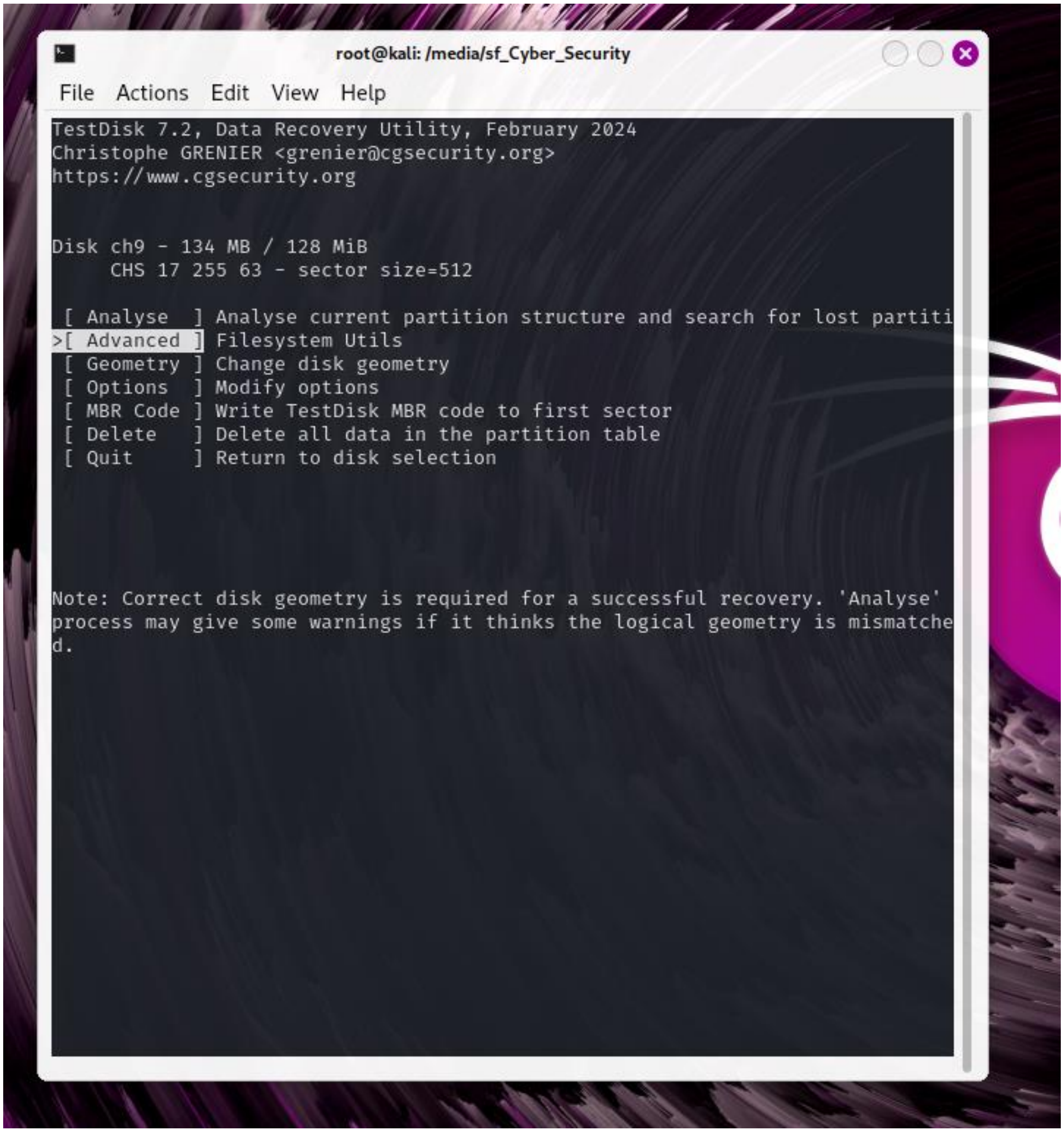
1. **Configure the Shared Folder:**
   - Navigate to **Settings → Shared Folders → Add Shared**.
   - Under **Folder Path**, select **Other**.
   - Choose the desired folder to share.
   - 

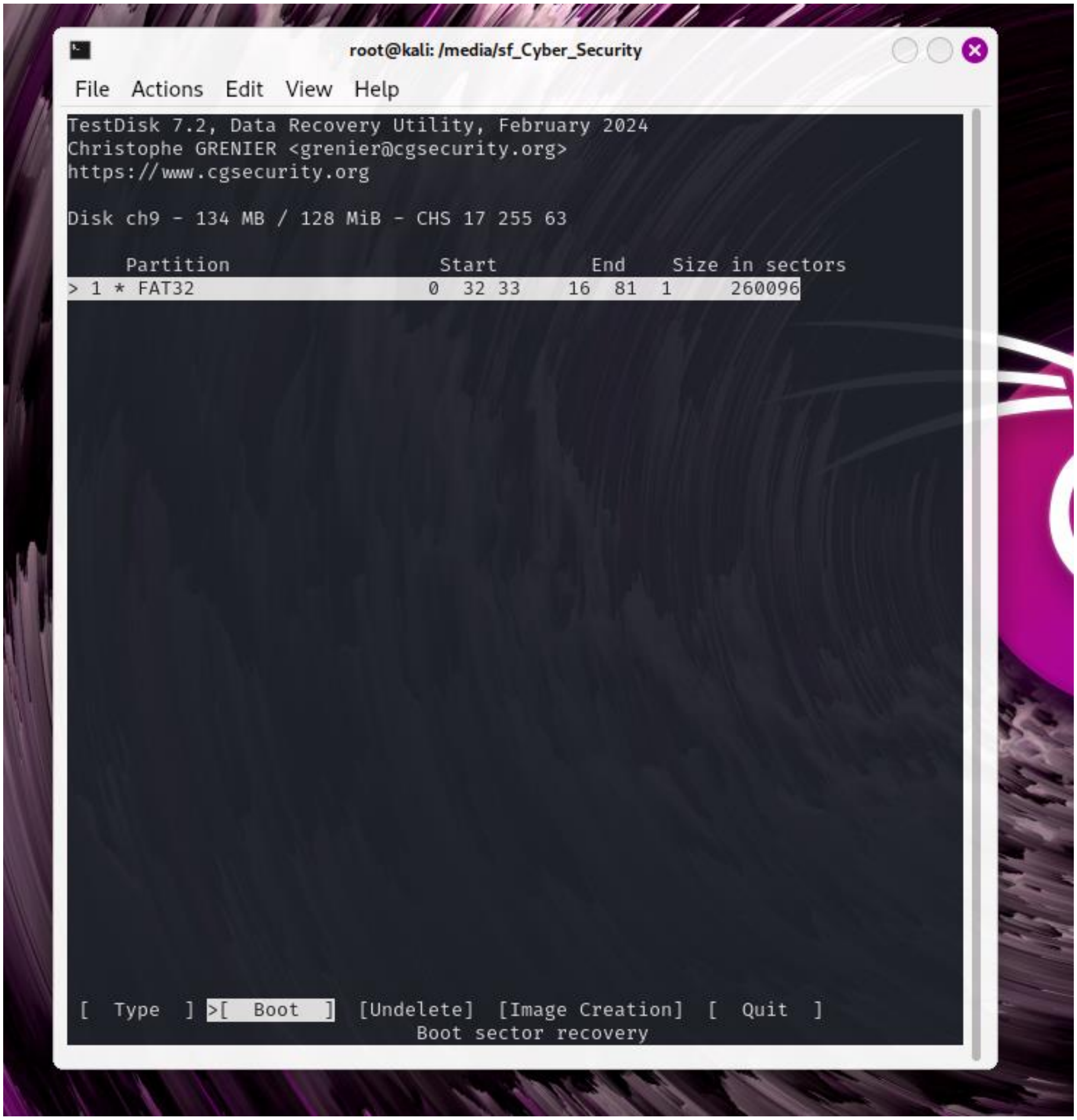
2. **Verify Shared Folder in Terminal:**
   - Gain root access by typing: sudo -i.
   - Navigate to the shared folder using: cd /media/sf_JU (*sf* represents "shared folder," and *JU* is the folder name).
   - Confirm that the target file exists in the shared folder.



3. **Recover Deleted Files:**
   - Use the command testdisk ch9, and follow these steps:
     - Select **Process → Intel → Advanced → Boot → List → Files**.
     - Locate the red line, navigate to it, and type C twice to copy the deleted file to the target folder (*JU*).


root@kali: /media/sf_Cyber_Security
File Actions Edit View Help
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.
Select a media and choose 'Proceed' using arrow keys:
>Disk ch9 - 134 MB / 128 MiB
>[Proceed] [Quit]
Note:
Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has an incorrect size, check HD jumper settings and BI
detection, and install the latest OS patches and disk drivers.

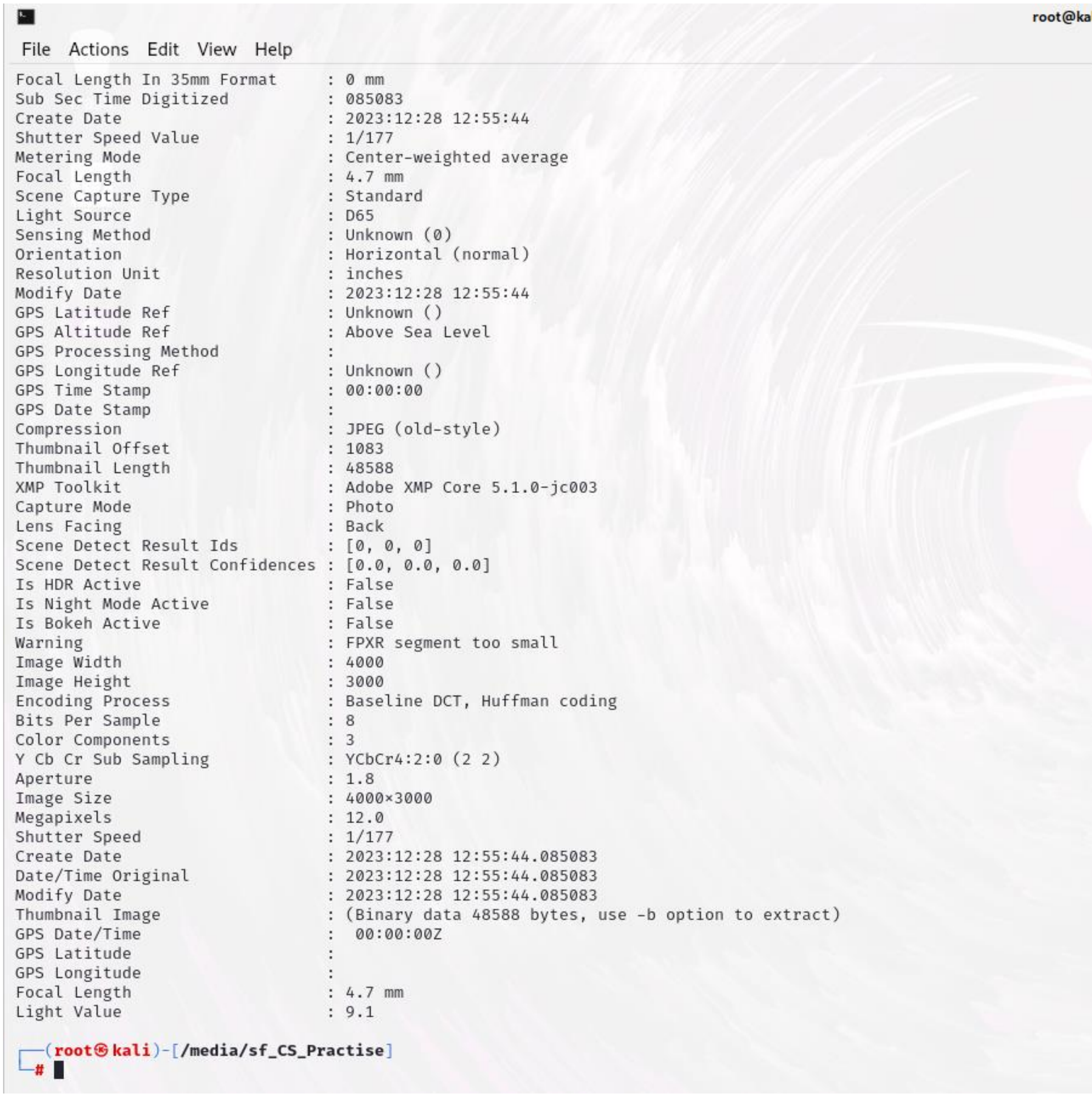

root@kali: /media/sf_Cyber_Security
File Actions Edit View Help
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
Disk ch9 - 134 MB / 128 MiB
CHS 17 255 63 - sector size=512
[Analyse] Analyse current partition structure and search for lost partiti
>[Advanced] Filesystem Utils
[Geometry] Change disk geometry
[Options] Modify options
[MBR Code] Write TestDisk MBR code to first sector
[Delete] Delete all data in the partition table
[Quit] Return to disk selection
Note: Correct disk geometry is required for a successful recovery. 'Analyse'
process may give some warnings if it thinks the logical geometry is mismatche
d.



4. **Extract Files:**
   - Open the target folder (*JU*), where a new folder named "File" will be created.
   - Change the file extension to ZIP (without compressing).
   - Extract all files from the ZIP folder.
5. **Analyze Picture Metadata:**
   - Navigate to the folder containing the pictures using cd.
   - Use the command: exiftool picture_name to view detailed metadata of the image.

```
┌──(root㉿kali)-[/media/sf_CS_Practise]
└─# exiftool IMG_20231228_125543.jpg
ExifTool Version Number         : 12.76
File Name                       : IMG_20231228_125543.jpg
Directory                       : .
File Size                       : 4.5 MB
File Modification Date/Time     : 2024:08:30 11:34:55-04:00
File Access Date/Time           : 2024:12:03 13:15:27-05:00
File Inode Change Date/Time     : 2024:08:30 11:34:55-04:00
File Permissions                : -rwxrwx---
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
Exif Byte Order                 : Big-endian (Motorola, MM)
Y Resolution                    : 72
X Resolution                    : 72
Camera Model Name               : KB2005
Make                            : OnePlus
Y Cb Cr Positioning             : Centered
Exif Version                    : 0220
Aperture Value                  : 1.7
Scene Type                      : Unknown (0)
Exposure Compensation           : 0
Exposure Program                : Not Defined
Color Space                     : sRGB
Max Aperture Value              : 1.7
Exif Image Height               : 3000
Brightness Value                : 0
Date/Time Original              : 2023:12:28 12:55:44
Flashpix Version                : 0100
Sub Sec Time Original           : 085083
White Balance                   : Auto
Interoperability Index          : R98 - DCF basic file (sRGB)
Exposure Mode                   : Auto
Exposure Time                   : 1/177
Flash                           : Off, Did not fire
Sub Sec Time                    : 085083
F Number                        : 1.8
ISO                             : 100
Exif Image Width                : 4000
Components Configuration        : Y, Cb, Cr, -
Focal Length In 35mm Format     : 0 mm
Sub Sec Time Digitized          : 085083
Create Date                     : 2023:12:28 12:55:44
Shutter Speed Value             : 1/177
Metering Mode                   : Center-weighted average
Focal Length                    : 4.7 mm
Scene Capture Type              : Standard
Light Source                    : D65
Sensing Method                  : Unknown (0)
Orientation                     : Horizontal (normal)
Resolution Unit                 : inches
```

6. **Locate the Picture:**
    - Obtain the GPS Latitude and Longitude from the metadata.
    - Visit GPS Coordinates and input the coordinates to determine the location.



```
File  Actions  Edit  View  Help
Focal Length In 35mm Format     : 0 mm
Sub Sec Time Digitized          : 085083
Create Date                     : 2023:12:28 12:55:44
Shutter Speed Value             : 1/177
Metering Mode                   : Center-weighted average
Focal Length                    : 4.7 mm
Scene Capture Type              : Standard
Light Source                    : D65
Sensing Method                  : Unknown (0)
Orientation                     : Horizontal (normal)
Resolution Unit                 : inches
Modify Date                     : 2023:12:28 12:55:44
GPS Latitude Ref                : Unknown ()
GPS Altitude Ref                : Above Sea Level
GPS Processing Method           :
GPS Longitude Ref               : Unknown ()
GPS Time Stamp                  : 00:00:00
GPS Date Stamp                  :
Compression                     : JPEG (old-style)
Thumbnail Offset                : 1083
Thumbnail Length                : 48588
XMP Toolkit                     : Adobe XMP Core 5.1.0-jc003
Capture Mode                    : Photo
Lens Facing                     : Back
Scene Detect Result Ids         : [0, 0, 0]
Scene Detect Result Confidences : [0.0, 0.0, 0.0]
Is HDR Active                   : False
Is Night Mode Active            : False
Is Bokeh Active                 : False
Warning                         : FPXR segment too small
Image Width                     : 4000
Image Height                    : 3000
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:2:0 (2 2)
Aperture                        : 1.8
Image Size                      : 4000x3000
Megapixels                      : 12.0
Shutter Speed                   : 1/177
Create Date                     : 2023:12:28 12:55:44.085083
Date/Time Original              : 2023:12:28 12:55:44.085083
Modify Date                     : 2023:12:28 12:55:44.085083
Thumbnail Image                 : (Binary data 48588 bytes, use -b option to extract)
GPS Date/Time                   :  00:00:00Z
GPS Latitude                    :
GPS Longitude                   :
Focal Length                    : 4.7 mm
Light Value                     : 9.1

┌──(root㉿kali)-[/media/sf_CS_Practise]
└─#
```

This assignment demonstrates how to set up a shared folder, recover and analyze files, and extract location data from images.